# Report 3: Bonus Task

# Streamlit App

Web application is designed to aid in the classification and evaluation of cybersecurity incidents using the **Gemini 1.5** and **Gemini 2.0** language models. It enables users to classify new incidents, evaluate the model's performance, explore and annotate incident data, and perform error analysis. The app is built using **Streamlit**, providing an interactive and user-friendly interface.

## Workflow of the Application

The app operates with six main pages, each designed to handle different tasks related to the classification of incidents and model evaluation:

# Page1:Classify Incident

- **Objective**: Classify new cybersecurity incidents into predefined categories using Gemini 1.5 or 2.0.
- **Workflow**:
    1. **Model Selection**: User selects which model version (1.5 or 2.0) to use for classification.
    2. **Incident Input**: User enters the incident description in a text area.
    3. **LLM Query**: The app sends the incident description to the chosen model via **LangChain** and **Google Generative AI** API for classification.
    4. **Result Parsing**: The model returns a JSON list of categories, which the app parses and displays to the user.
    5. **Output**: The predicted categories are displayed on the UI for the user.

# Page2:Model Evaluation

- **Objective**: Evaluate and compare the performance of Gemini 1.5 and 2.0 models on historical data.
- **Workflow**:
    1. **Model Selection**: User selects which model (1.5 or 2.0) to evaluate.
    2. **Metrics Calculation**: The app computes evaluation metrics like Micro F1, Macro F1, and Jaccard Index (Micro & Macro) using **sklearn**.
    3. **Visualization**: Displays:
        - Bar chart for per-label F1 scores using **Altair**.
        - Full classification report with precision, recall, F1 scores.

## Page3:Data Explorer

- **Objective**: Explore the dataset of incidents and apply various filters.
- **Workflow**:
    1. **Search Function**: User can search incidents by title, summary, or keyword.
    2. **Filtering**: Filters incidents by:
        - True category labels.
        - Match or mismatch between predictions of Gemini 1.5 and 2.0.
    3. **Output**:
        - Displays filtered incidents with their true categories and model predictions.
        - Displays the distribution of incident categories as a bar chart.
        - Shows a word frequency table for incident summaries.

## Page4: LLM Comparison

- **Objective**: Compare the predictions of Gemini 1.5 and 2.0 and evaluate their accuracy.
- **Workflow**:
    1. **Mismatch Detection**: Finds incidents where predictions from the two models differ.
    2. **Evaluation**: Displays the number of correct and incorrect predictions for both models.
    3. **Comparison Table**: Shows a detailed table of predicted categories, true labels, and model correctness for both Gemini 1.5 and 2.0.

## Page5: Annotation Tool

- **Objective**: Allow manual correction of model predictions.
- **Workflow**:
    1. **Incident Selection**: User selects an incident by index.
    2. **Update Prediction**: User enters the correct predicted categories for Gemini 2.0.
    3. **Save Updates**: On clicking "Update," the corrected predictions are saved to a new CSV (Annotation_tool_output.csv).

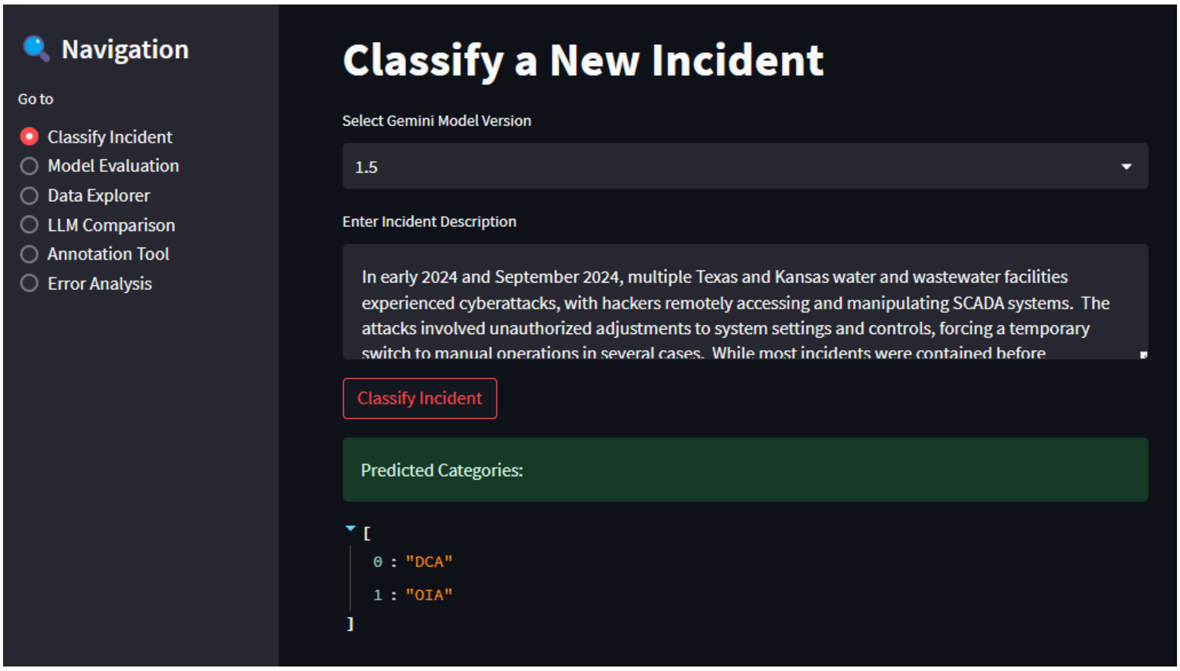
## Page6: Error Analysis

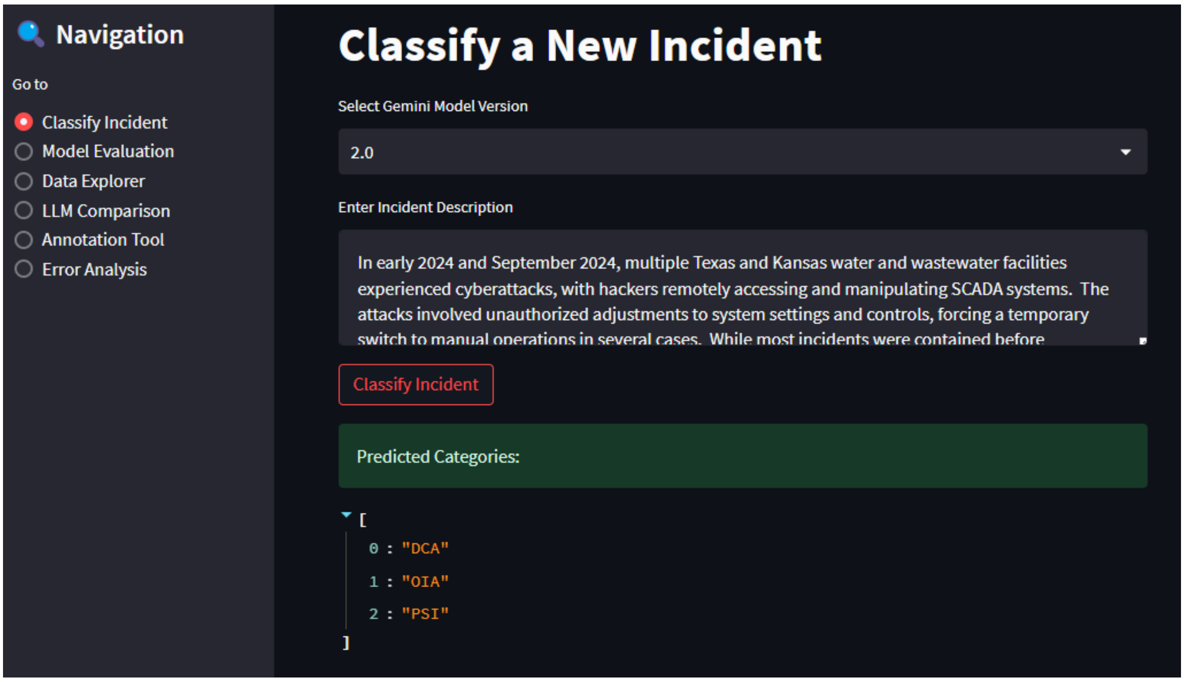
- **Objective**: Visualize and analyze the errors made by the model during classification.
- **Workflow**:
    1. **Error Analysis**: Displays a bar plot showing the error count for each label.

2. **Sample Misclassifications**: Displays a table of incidents where the model made incorrect predictions.

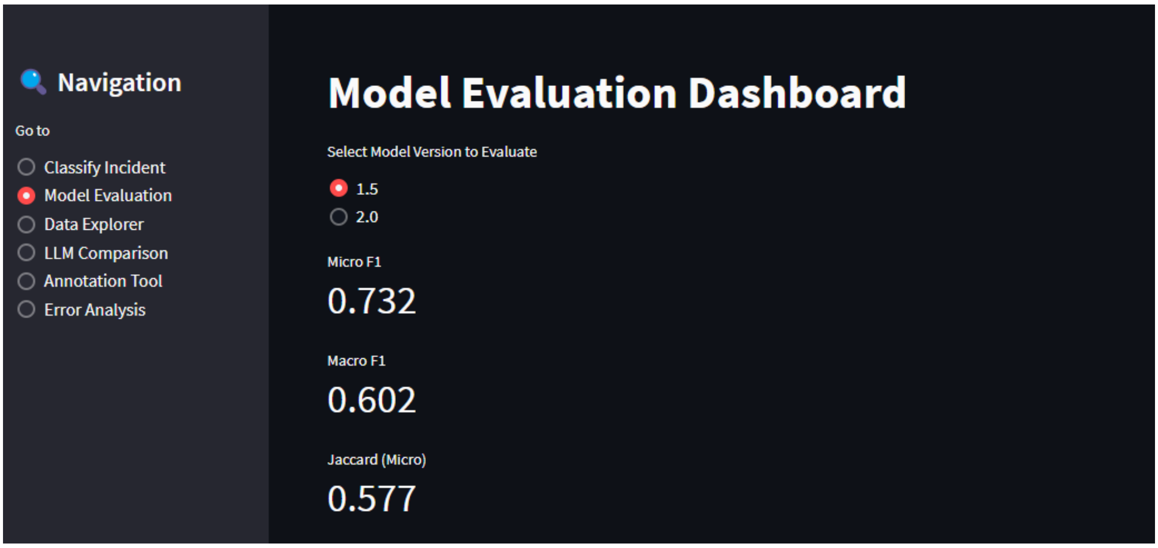
## Output:

## Page1:

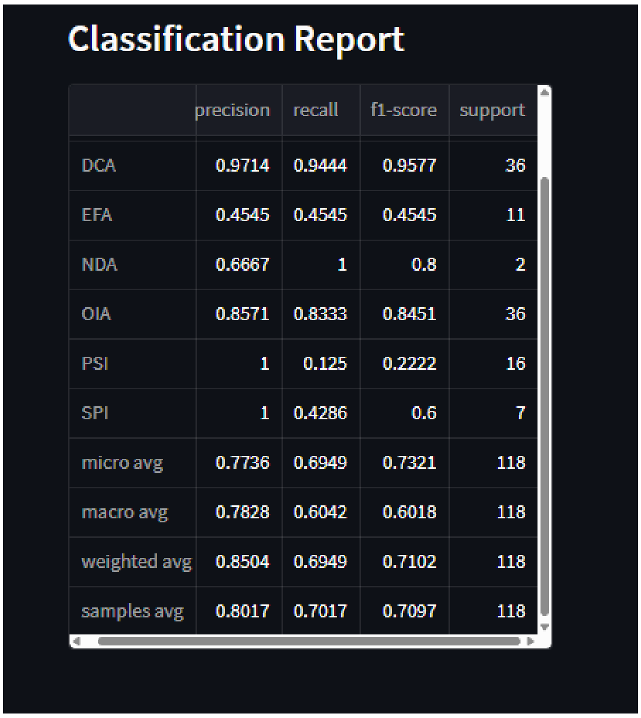


## Page2:







### Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| DCA | 0.9714 | 0.9444 | 0.9577 | 36 |
| EFA | 0.4545 | 0.4545 | 0.4545 | 11 |
| NDA | 0.6667 | 1 | 0.8 | 2 |
| OIA | 0.8571 | 0.8333 | 0.8451 | 36 |
| PSI | 1 | 0.125 | 0.2222 | 16 |
| SPI | 1 | 0.4286 | 0.6 | 7 |
| micro avg | 0.7736 | 0.6949 | 0.7321 | 118 |
| macro avg | 0.7828 | 0.6042 | 0.6018 | 118 |
| weighted avg | 0.8504 | 0.6949 | 0.7102 | 118 |
| samples avg | 0.8017 | 0.7017 | 0.7097 | 118 |

**Page 3:**

## LLM Prediction Match Filter

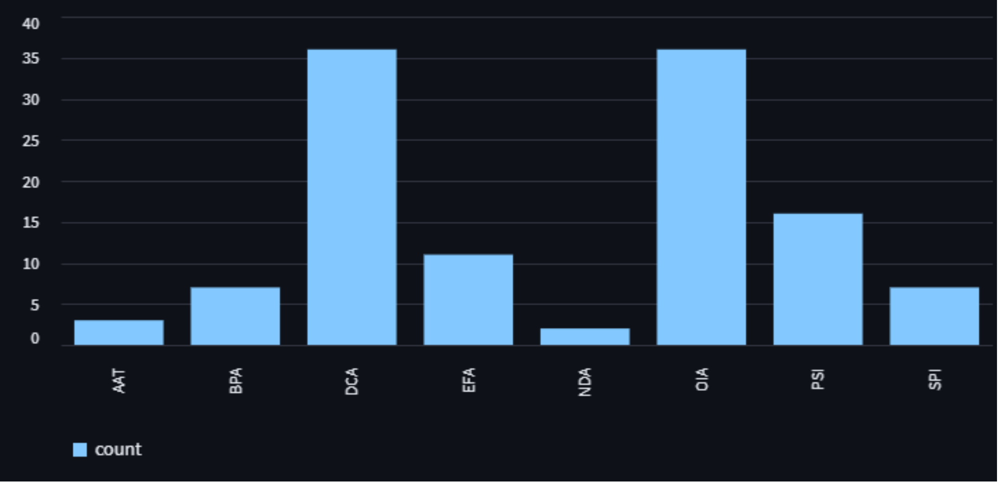
Filter by Prediction Match

- (●) All
- ( ) Only Matches
- ( ) Only Mismatches

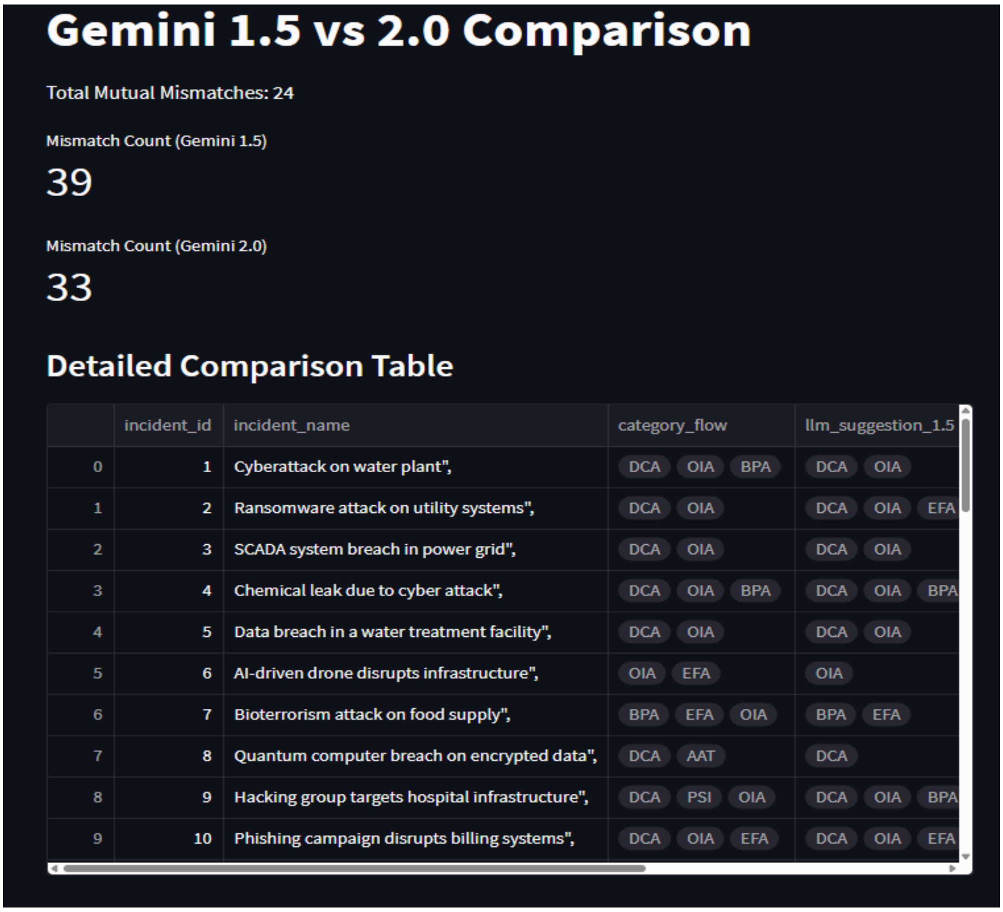
## Incidents Table

| | incident_id | incident_name | incident_summary |
|---|---|---|---|
| 0 | 1 | Cyberattack on water plant", | In 2024, several water and wastewater faciliti |
| 1 | 2 | Ransomware attack on utility systems", | Ransomware attacks, where malicious softw |
| 2 | 3 | SCADA system breach in power grid", | Cyberattacks targeting critical infrastructure, |
| 3 | 4 | Chemical leak due to cyber attack", | A recent audit revealed that many US chemic |
| 4 | 5 | Data breach in a water treatment facility", | In September 2024, the Arkansas City, Kansa |
| 5 | 6 | AI-driven drone disrupts infrastructure", | AI-controlled drones could be used in coordi |
| 6 | 7 | Bioterrorism attack on food supply", | An attack on the food supply, likely detected |
| 7 | 8 | Quantum computer breach on encrypted data", | A potential cybersecurity threat involves atta |
| 8 | 9 | Hacking group targets hospital infrastructure", | In July 2024, the U.S. Department of Justice a |
| 9 | 10 | Phishing campaign disrupts billing systems", | American Water experienced a cybersecurity |

## Label Distribution



## Page 4:



# Gemini 1.5 vs 2.0 Comparison

Total Mutual Mismatches: 24

Mismatch Count (Gemini 1.5)

39

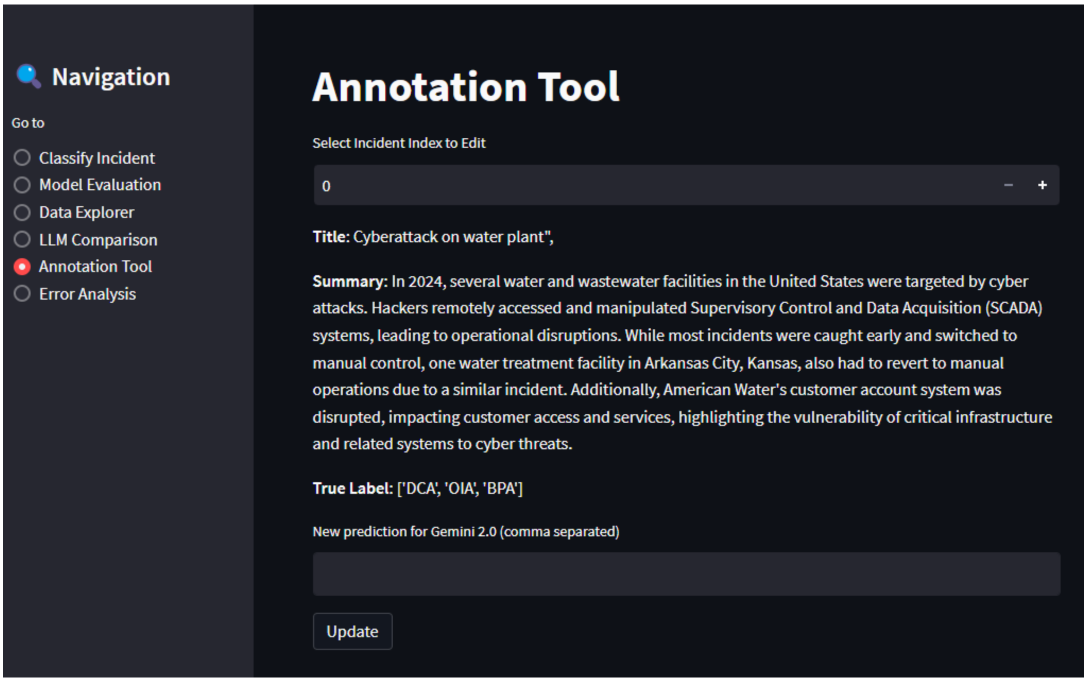
Mismatch Count (Gemini 2.0)

33

## Detailed Comparison Table

| | incident_id | incident_name | category_flow | llm_suggestion_1.5 |
|---|---|---|---|---|
| 0 | 1 | Cyberattack on water plant", | DCA OIA BPA | DCA OIA |
| 1 | 2 | Ransomware attack on utility systems", | DCA OIA | DCA OIA EFA |
| 2 | 3 | SCADA system breach in power grid", | DCA OIA | DCA OIA |
| 3 | 4 | Chemical leak due to cyber attack", | DCA OIA BPA | DCA OIA BPA |
| 4 | 5 | Data breach in a water treatment facility", | DCA OIA | DCA OIA |
| 5 | 6 | AI-driven drone disrupts infrastructure", | OIA EFA | OIA |
| 6 | 7 | Bioterrorism attack on food supply", | BPA EFA OIA | BPA EFA |
| 7 | 8 | Quantum computer breach on encrypted data", | DCA AAT | DCA |
| 8 | 9 | Hacking group targets hospital infrastructure", | DCA PSI OIA | DCA OIA BPA |
| 9 | 10 | Phishing campaign disrupts billing systems", | DCA OIA EFA | DCA OIA EFA |

## Page 5:





# Annotation Tool

Select Incident Index to Edit

0

**Title:** Cyberattack on water plant",

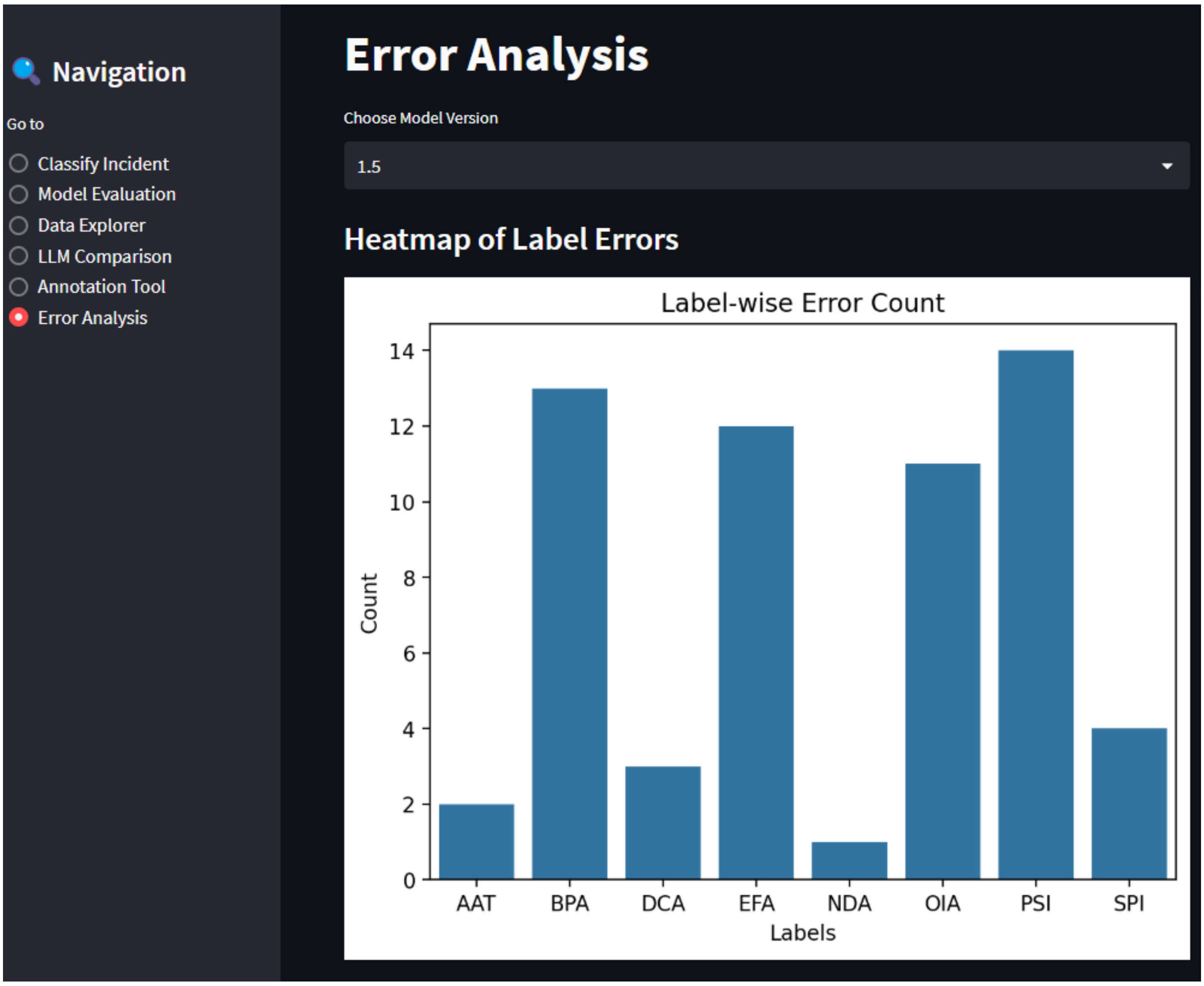
**Summary:** In 2024, several water and wastewater facilities in the United States were targeted by cyber attacks. Hackers remotely accessed and manipulated Supervisory Control and Data Acquisition (SCADA) systems, leading to operational disruptions. While most incidents were caught early and switched to manual control, one water treatment facility in Arkansas City, Kansas, also had to revert to manual operations due to a similar incident. Additionally, American Water's customer account system was disrupted, impacting customer access and services, highlighting the vulnerability of critical infrastructure and related systems to cyber threats.

**True Label:** ['DCA', 'OIA', 'BPA']

New prediction for Gemini 2.0 (comma separated)

Update

## Page 6:




## Sample Misclassifications

| | incident_name | category_flow | llm_suggestion_1.5 |
|---|---|---|---|
| 0 | Cyberattack on water plant", | DCA OIA BPA | DCA OIA |
| 1 | Ransomware attack on utility systems", | DCA OIA | DCA OIA EFA |
| 5 | AI-driven drone disrupts infrastructure", | OIA EFA | OIA |
| 6 | Bioterrorism attack on food supply", | BPA EFA OIA | BPA EFA |
| 7 | Quantum computer breach on encrypted data", | DCA AAT | DCA |
| 8 | Hacking group targets hospital infrastructure", | DCA PSI OIA | DCA OIA BPA |
| 9 | Phishing campaign disrupts billing systems", | DCA OIA EFA | DCA OIA EFA BPA |
| 10 | Power outage caused by cyber attack", | DCA OIA PSI | OIA |
| 11 | Confirmed malware on oil pipeline sensors", | DCA OIA EFA | DCA OIA |
| 12 | Attack on agricultural database", | DCA BPA | DCA OIA BPA EFA |

## Design & Structure Rationale

1. **Modular Page Design (Sidebar Navigation):**
   Cleanly separates core functionalities—classification, evaluation, exploration, and annotation—into independent modules for focused workflows and easier debugging.
2. **Efficient Data Handling:**
   Uses @st.cache_data to efficiently load and preprocess CSV data once, while early binarization of multi-label categories enables fast metric computation and evaluation.
3. **LLM Integration (Classify Incident):**
   Provides structured prompts with strict response formats and fallback parsing to ensure consistent and reliable outputs from Gemini LLMs across versions.
4. **Visual & Metric-Driven Evaluation (Model Evaluation, LLM Comparison):**
   Leverages micro/macro F1, Jaccard index, and classification reports alongside Altair/Seaborn visualizations to offer both granular and aggregated performance insights.
5. **Exploratory Features (Data Explorer, Annotation Tool):**
   Enables intuitive filtering, mismatch analysis, and human-in-the-loop annotation for continuous dataset refinement and transparent model inspection.
6. **Error Analysis:**
   Delivers per-label error breakdowns and surfacing of misclassified samples via graphs and tables to identify and address model weaknesses or data issues.

**Note**: Check Readme file to understand how to run.